

# Measurement of $\Psi(2S)$ cross section from dimuon invariant mass

Statistical Methods and Analysis Techniques in Experimental Physics

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# Outline of the project

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  - choose an appropriate model;
  - minimize the **unbinned extended negative log-likelihood**;
  - test the goodness of fit: residuals,  $\chi^2$ , run test.

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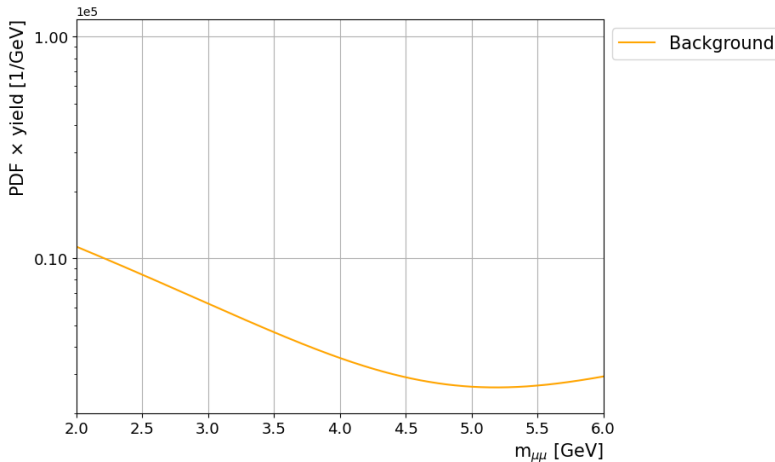
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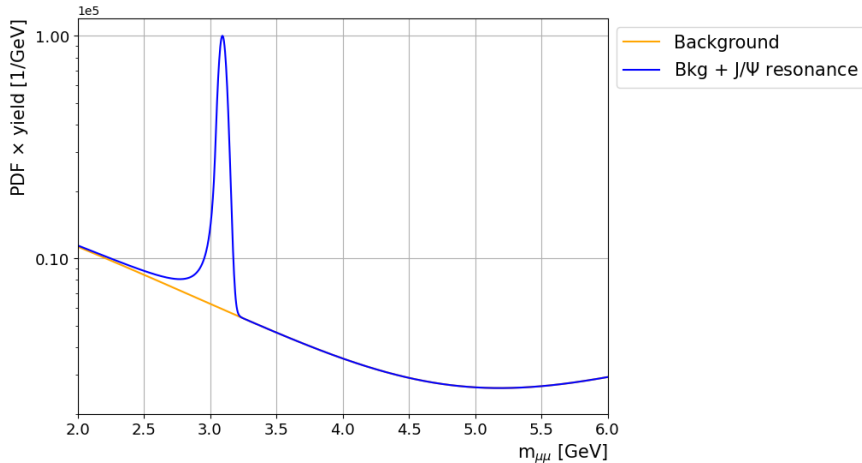
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4. Re-fit  $\sigma_{\Psi(2S)}$  using a low statistic dataset and set an **upper limit** using  $q_\mu$  as test statistic.

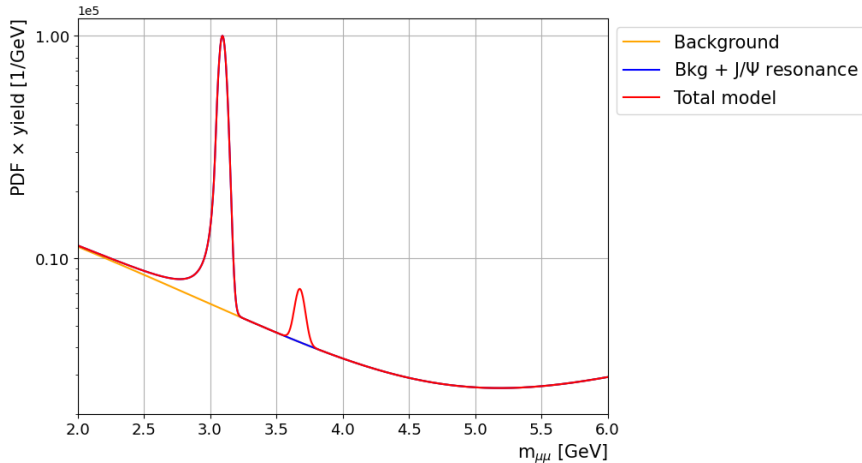
# Adopted model



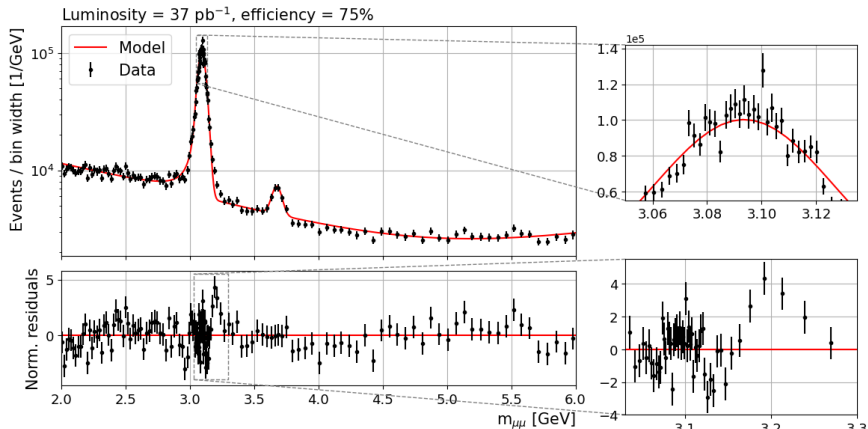
# Adopted model



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# Poorest fit: results



# Goodness of fit: Residuals and $\chi^2$ test

Number of bins: 156

Residuals in  $1\sigma$ : 56.4%

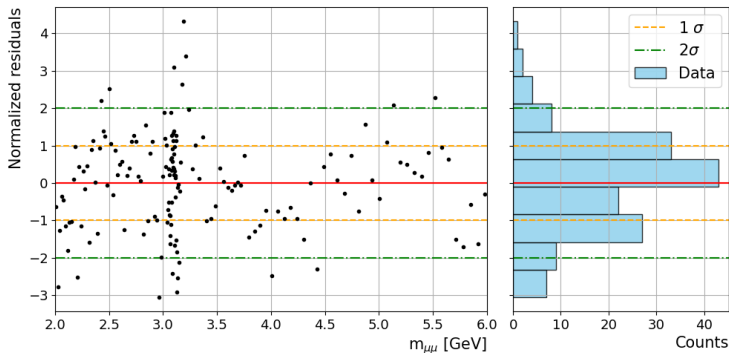
Residuals in  $2\sigma$ : 89.1%

$\chi^2/\text{ndf} = \mathbf{1.74}$ ,  $\text{ndf} = 144$

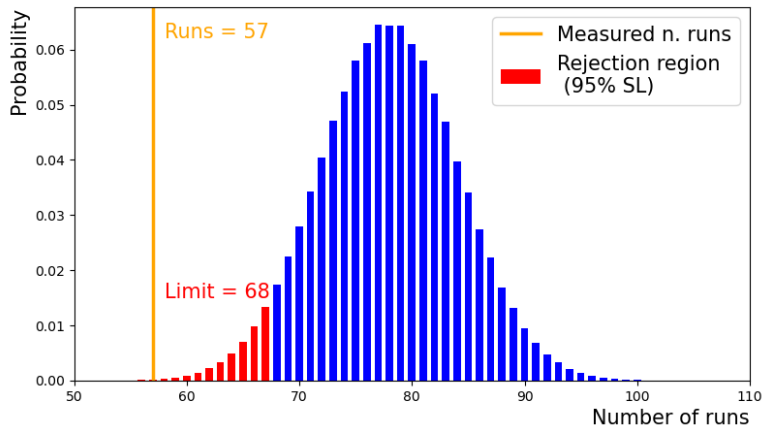
p-value =  $8.6 \times 10^{-8}$

One-tailed chi-squared test:

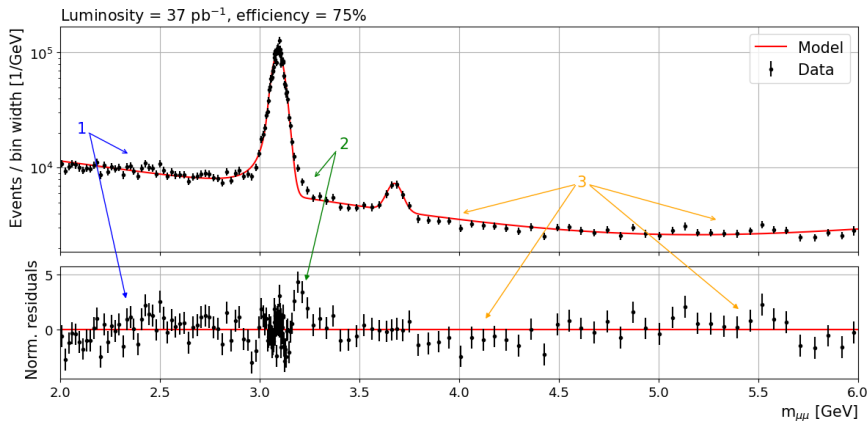
$\chi^2_{\text{max}}/\text{ndf}$  at 95% SL: **1.20**



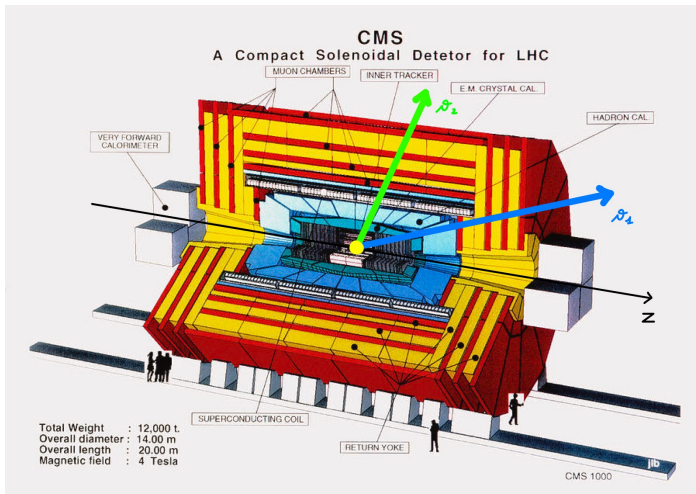
## Goodness of fit: Run test



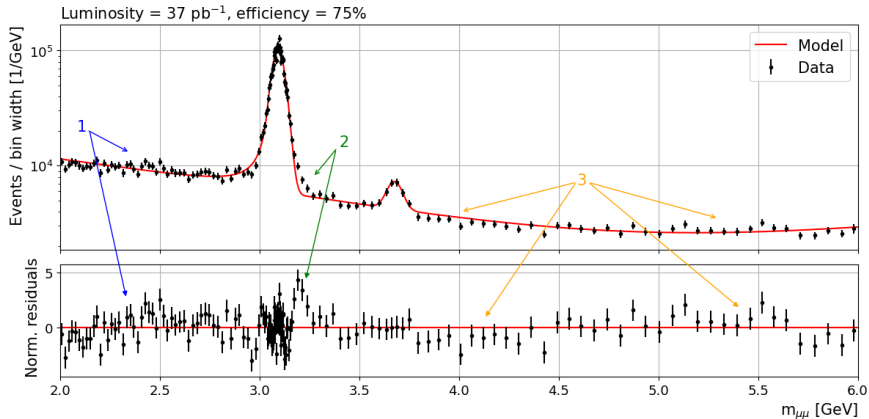
# Critical features of the fit



# Non-uniform resolution of CMS



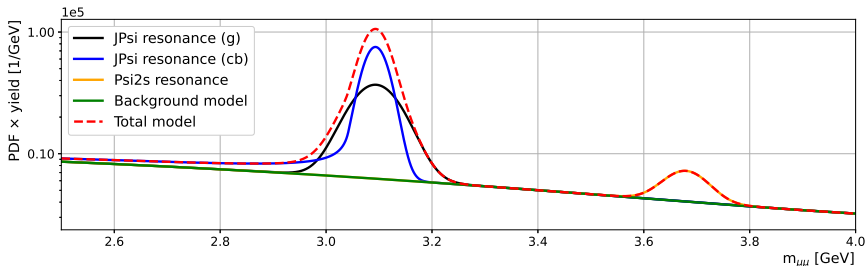
# Critical features of the fit



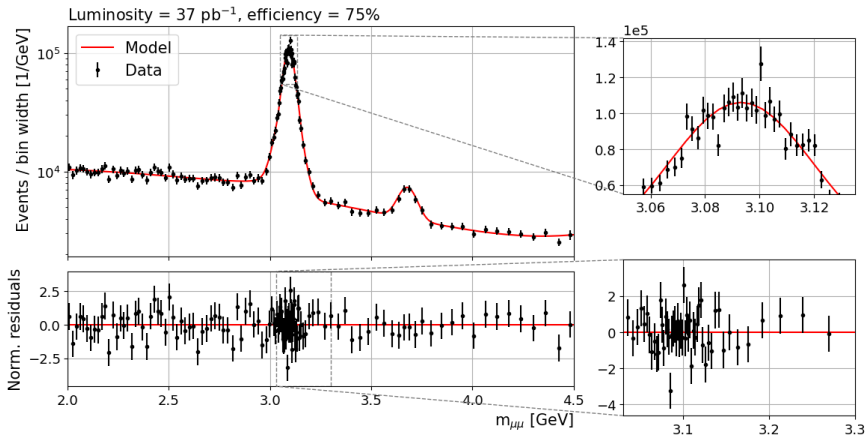
## Fit using high statistic dataset

In the range [2, 4.5] GeV:

- $J/\psi$  resonance: gaussian function + Crystall Ball (same  $\mu$ );
- $\psi(2S)$  resonance: gaussian function;
- Background: sum of Chebyshev polynomials up to order 3.



# Model and data comparison



# Goodness of fit: Residuals and $\chi^2$ test

Number of bins: 133

Residuals in  $1\sigma$ : 72.9%

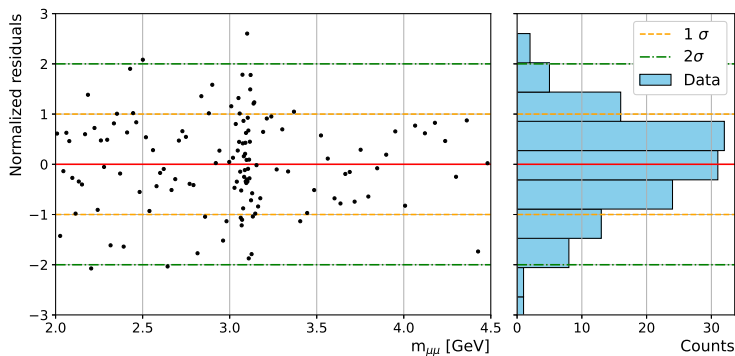
Residuals in  $2\sigma$ : 96.2%

$\chi^2/\text{ndf} = \mathbf{0.99}$ ,  $\text{ndf} = 119$

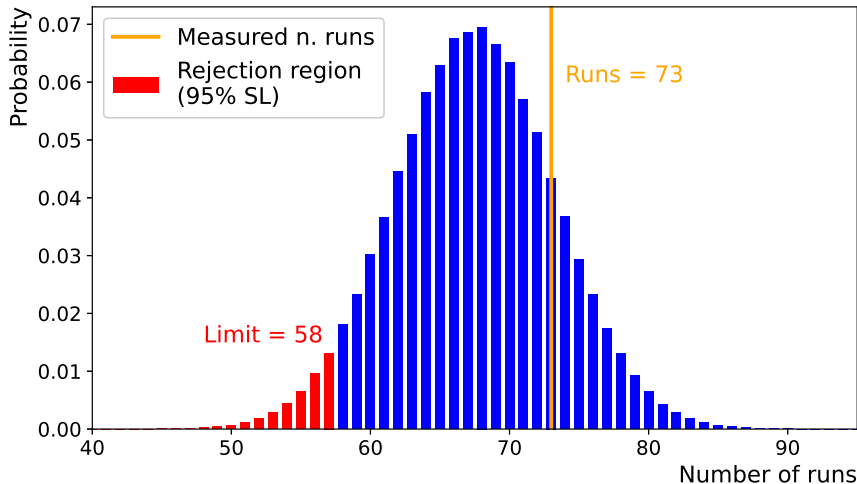
p-value = 0.46

One-tailed chi-squared test:

$\chi^2_{\text{max}}/\text{ndf}$  at 95% SL: **1.22**

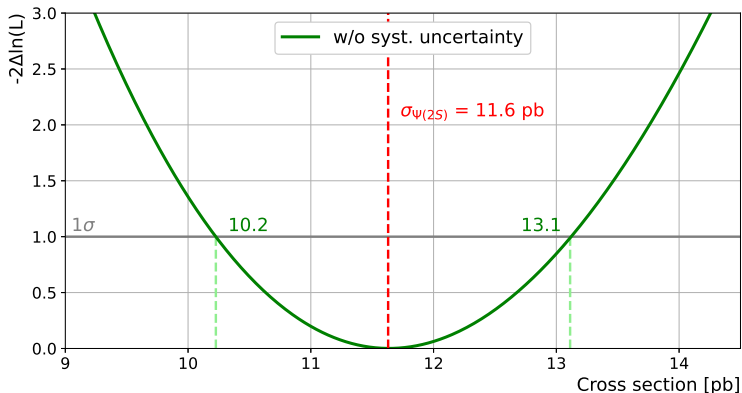


## Goodness of fit: Run test



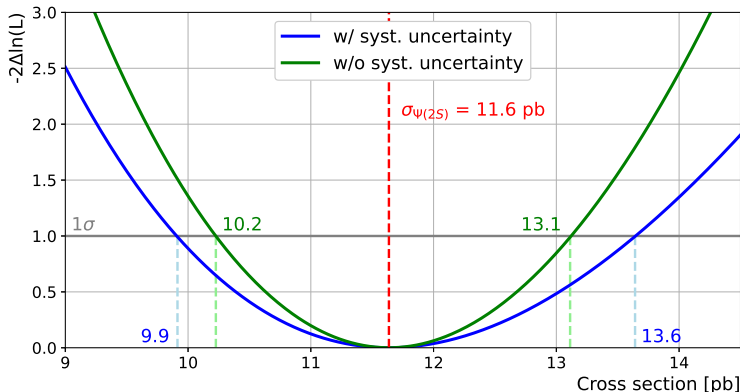
## 68% confidence interval: no systematic unc.

$$q_{\mu} = -2 \ln \left( \frac{L(\mu, \tilde{\theta})}{L(\hat{\mu}, \tilde{\theta})} \right) = \left( \frac{\mu - \hat{\mu}}{\sigma} \right)^2 + O(1/\sqrt{N}) \sim \chi^2(ndf = 1)$$



## 68% confidence interval: add systematic unc.

Fit signal efficiency constraining it through an additional Gaussian factor in the likelihood with 10% standard deviation.

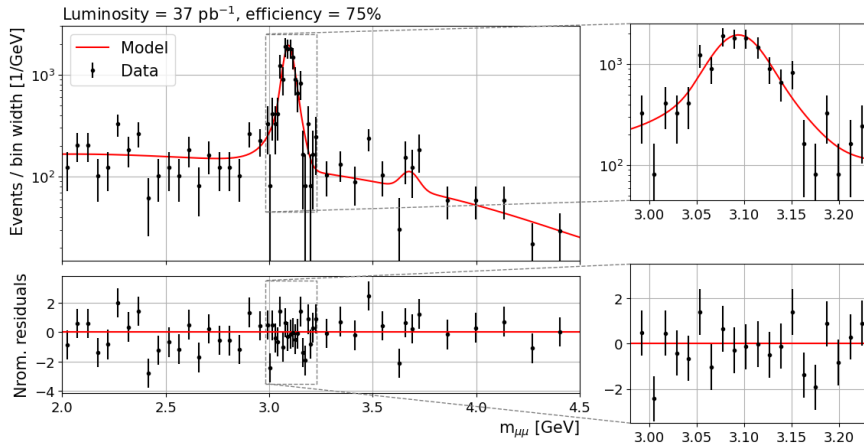


## Final cross section measurement

$$\sigma_{\psi(2S)} = 11.6^{+2.0}_{-1.7} \text{ pb}$$

(68% CI)

# Fit of the low statistic dataset

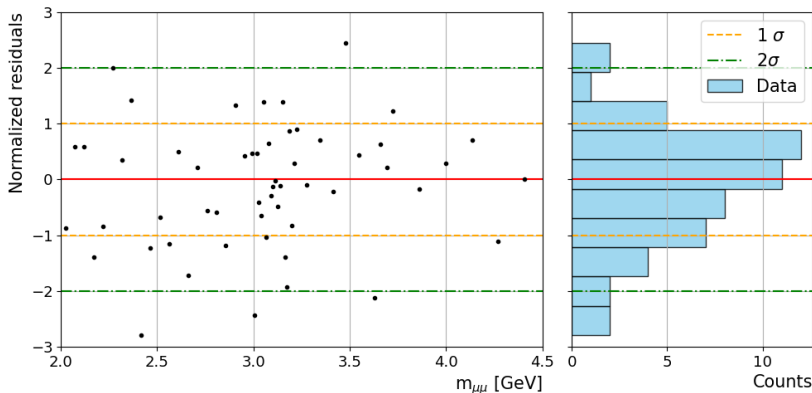


# Goodness of fit: Normalized residuals

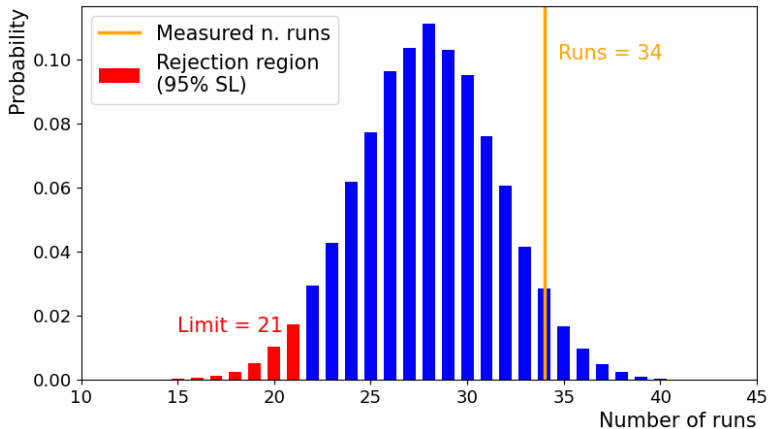
Number of bins: 54

Residuals in  $1\sigma$ : 64.8%

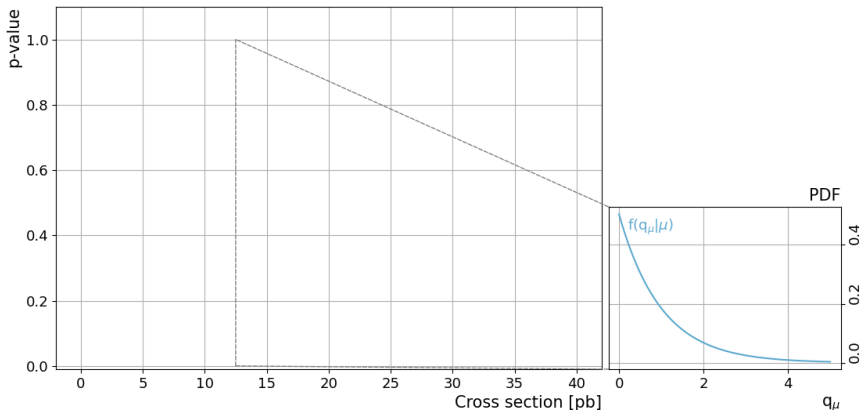
Residuals in  $2\sigma$ : 92.6%



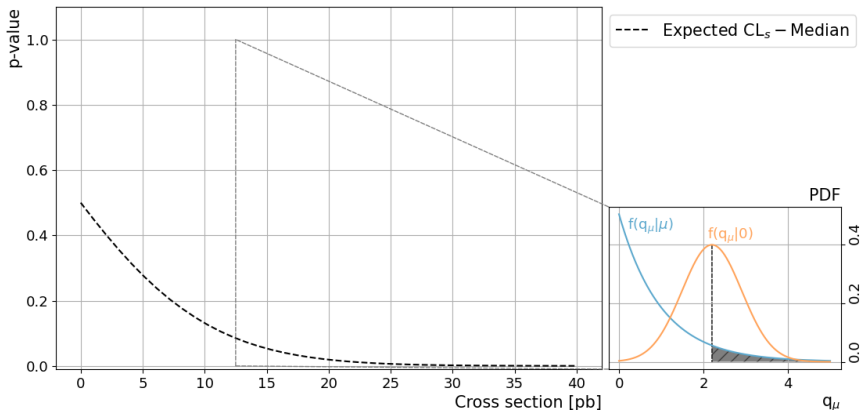
## Goodness of fit: Run test



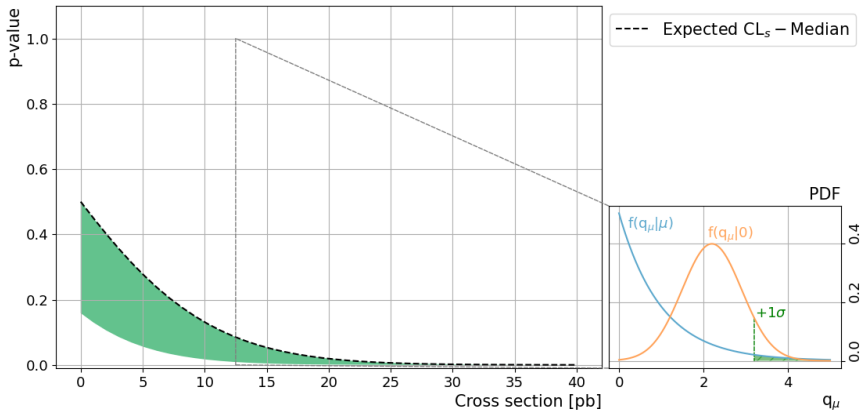
# Setting an upper limit to $\sigma_{\Psi(2S)}$



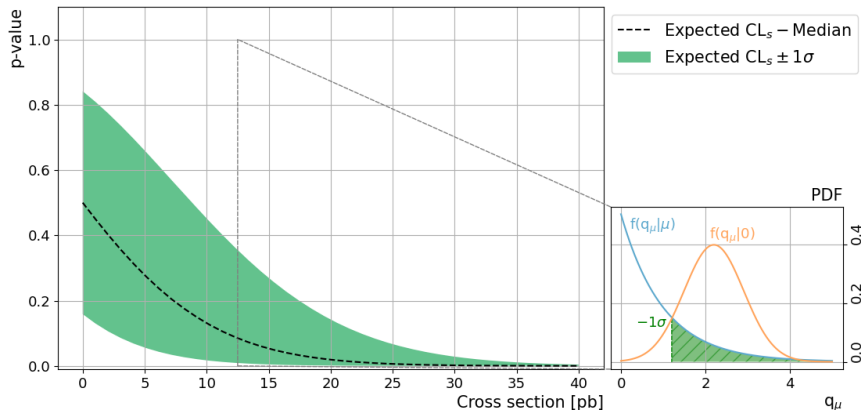
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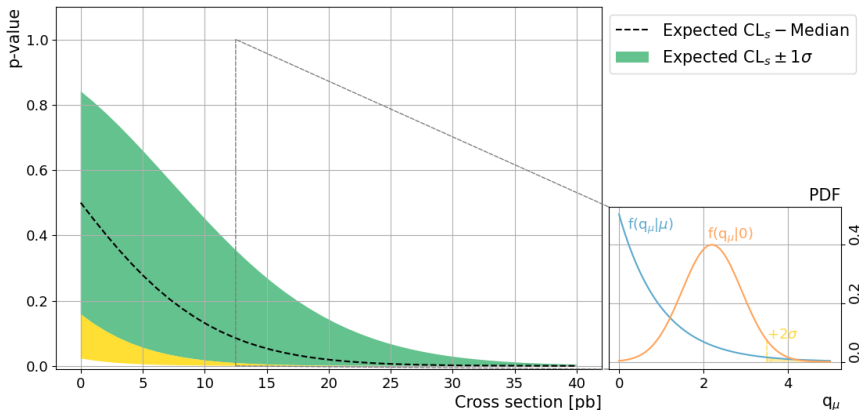
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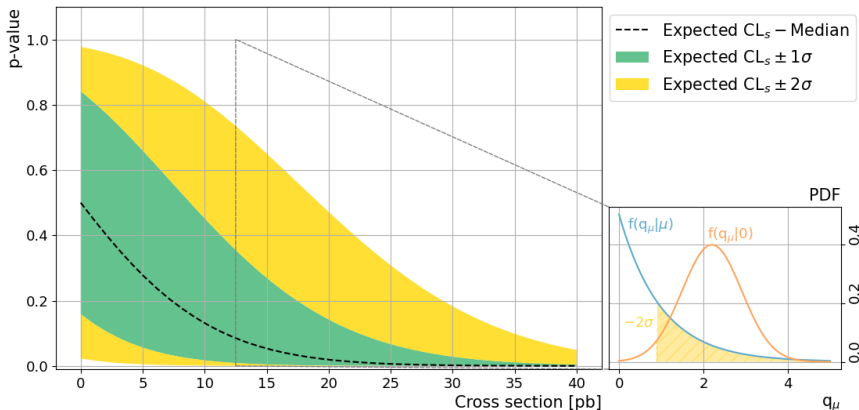
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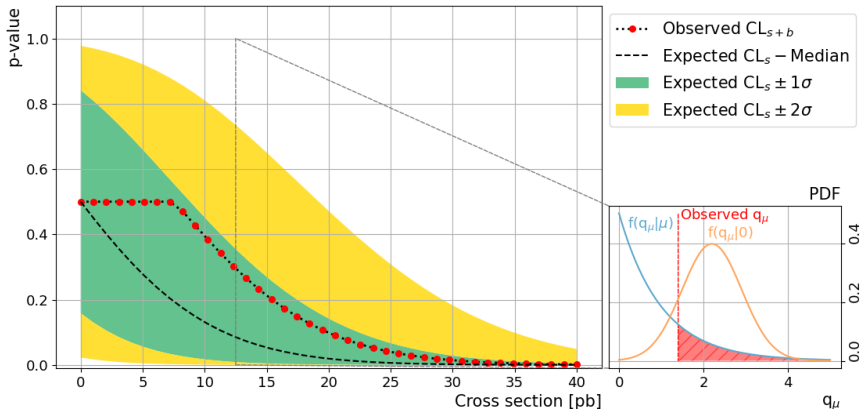
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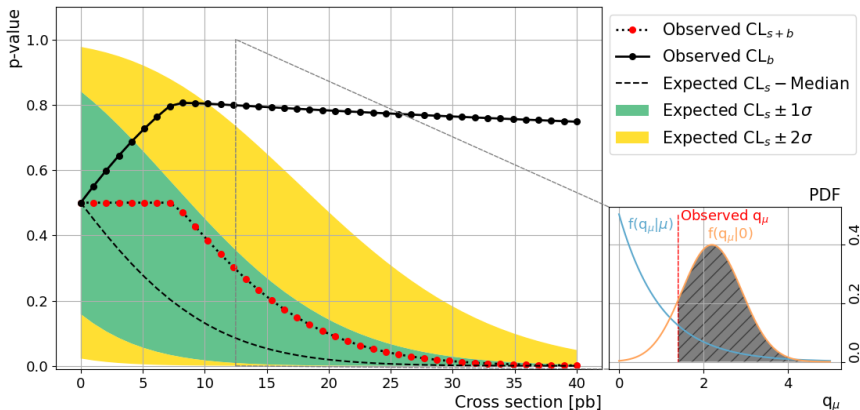
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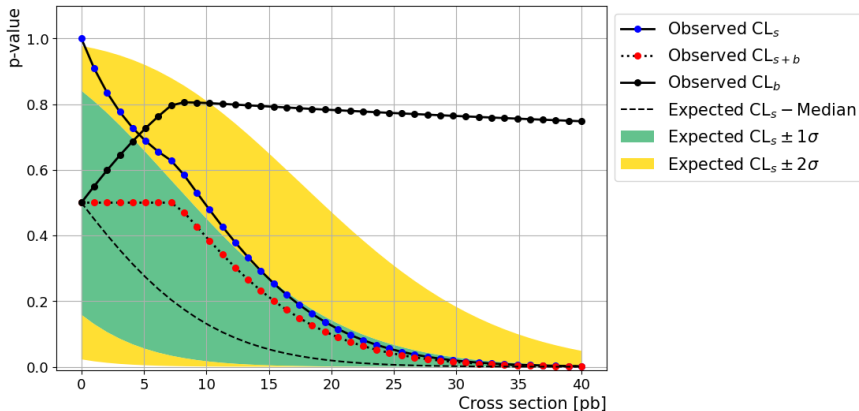
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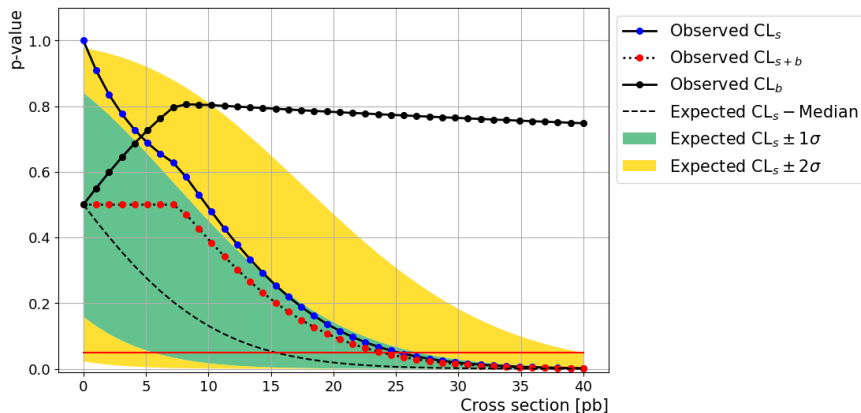
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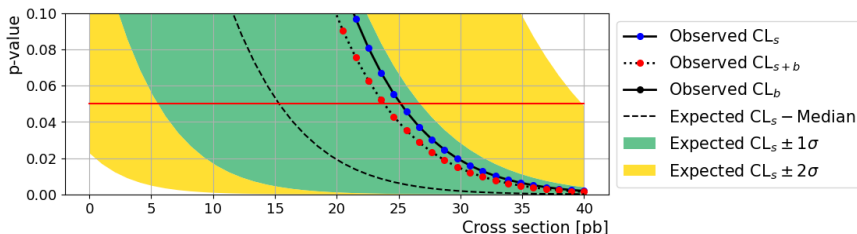
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Theoretical upper bound

Median: 15.3 pb

$\pm 1\sigma$  : [5.5, 26.7] pb

$\pm 2\sigma$  : [0, 39.8] pb

Observed upper bound

$CL_s$  : 25.2 pb

# Thanks for your attention!

Questions?

# CrystalBall for the $\Psi(2S)$ fit: table of parameters and results

| valid | converged | param at limit | edm    | approx. fmin (full   opt.) |
|-------|-----------|----------------|--------|----------------------------|
| False | True      | False          | 0.0013 | 23038.52   4102.636        |

| Parameters<br>name | value (rounded) | hesse       | at limit |
|--------------------|-----------------|-------------|----------|
| yield_jpsi         | 8820.51         | +/- 2.3e+02 | False    |
| xs_psi2s           | 12.3912         | +/- 10      | False    |
| yield_bkg          | 18858.2         | +/- 4.7e+02 | False    |
| mu_jpsi            | 3.09355         | +/- 0.00049 | False    |
| sigma_jpsi         | 0.0332748       | +/- 0.00051 | False    |
| alpha_jpsi         | 1.65117         | +/- 0.084   | False    |
| n_jpsi             | 1.3477          | +/- 0.28    | False    |
| mu_psi2s           | 3.67741         | +/- 0.013   | False    |
| sigma_psi2s        | 0.0383655       | +/- 0.015   | False    |
| alpha_psi2s        | 1.64387         | +/- 3.4     | False    |
| n_psi2s            | 1.06358         | +/- 1.3     | False    |
| a1                 | -0.748973       | +/- 0.016   | False    |
| a2                 | 0.336516        | +/- 0.015   | False    |
| a3                 | -0.0412214      | +/- 0.022   | False    |

# Worst fit: table of parameters and results

| valid | converged | param at limit | edm     | approx. fmin (full   opt.) |
|-------|-----------|----------------|---------|----------------------------|
| True  | True      | False          | 0.00034 | 23038.80   4103.522        |

| Parameters<br>name | value (rounded) | hesse       | at limit |
|--------------------|-----------------|-------------|----------|
| yield_jpsi         | 8764.55         | +/- 1.9e+02 | False    |
| xs_psi2s           | 10.4438         | +/- 1.3     | False    |
| yield_bkg          | 18971.9         | +/- 2.2e+02 | False    |
| mu_jpsi            | 3.09357         | +/- 0.0005  | False    |
| sigma_jpsi         | 0.0331908       | +/- 0.00052 | False    |
| alpha_jpsi         | 1.63126         | +/- 0.091   | False    |
| n_jpsi             | 1.43825         | +/- 0.29    | False    |
| mu_psi2s           | 3.67745         | +/- 0.0051  | False    |
| sigma_psi2s        | 0.0373381       | +/- 0.0051  | False    |
| a1                 | -0.749397       | +/- 0.015   | False    |
| a2                 | 0.333967        | +/- 0.012   | False    |
| a3                 | -0.0354694      | +/- 0.012   | False    |

# Worst run test: table of parameters and results

```
# Counting how many adjacent residuals have the same sign ***
```

```
# Counting the positive and negative residuals ***
```

```
Number of positive residuals: 87
```

```
Number of negative residuals: 69
```

```
Measured number of runs: 57
```

```
Expected number of runs: 78.0 with std = 6.1
```

```
#probability function of runs ***
```

```
Normalization: 0.9999999999999998
```

```
#Defining the rejection interval ***
```

```
Integrating up to R = 67: cumulative = 0.044
```

```
Given a significance level of 0.95, the critical region is  $R < R_{\min} = 68$ 
```

# Best fit: table of parameters and results

| valid | converged | param at limit | edm    | approx. fmin (full   opt.) |
|-------|-----------|----------------|--------|----------------------------|
| True  | True      | False          | 0.0003 | 9531.09   7076.038         |

| Parameters    |           |             |       |          |
|---------------|-----------|-------------|-------|----------|
| name          | value     | (rounded)   | hesse | at limit |
| yield_cb_jpsi | 5157.44   | +/- 5.2e+02 | False |          |
| yield_g_jpsi  | 3675.34   | +/- 5.5e+02 | False |          |
| xs_psi2s      | 11.6298   | +/- 1.4     | False |          |
| yield_bkg     | 14716.1   | +/- 3.5e+02 | False |          |
| mu_jpsi       | 3.09319   | +/- 0.00047 | False |          |
| sigma_cb_jpsi | 0.0251248 | +/- 0.0012  | False |          |
| alpha_jpsi    | 2.02748   | +/- 0.17    | False |          |
| n_jpsi        | 0.659364  | +/- 0.28    | False |          |
| sigma_g_jpsi  | 0.0477166 | +/- 0.0026  | False |          |
| mu_psi2s      | 3.67864   | +/- 0.0051  | False |          |
| sigma_psi2s   | 0.0404231 | +/- 0.0054  | False |          |
| a1            | -0.651518 | +/- 0.02    | False |          |
| a2            | 0.071954  | +/- 0.016   | False |          |
| a3            | 0.0584168 | +/- 0.017   | False |          |

# Best run test: table of parameters and results

```
# Counting how many adjacent residuals have the same sign ●●●
```

```
# Counting the positive and negative residuals ●●●
```

```
Number of positive residuals: 68
```

```
Number of negative residuals: 65
```

```
Measured number of runs: 73
```

```
Expected number of runs: 67.5 with std = 5.7
```

```
#probability function of runs ●●●
```

```
Normalization: 1.0
```

```
#Defining the rejection interval ●●●
```

```
Integrating up to  $R = 58$ : cumulative = 0.059
```

```
Given a significance level of 0.95, the critical region is  $R < R_{\min} = 58$ 
```

```
# Plotting the probability distribution of the runs number and Hypothesis Testing ●●●
```

# LS fit: table of parameters and results

| valid | converged | param at limit | edm     | approx. fmin (full   opt.) |
|-------|-----------|----------------|---------|----------------------------|
| True  | True      | False          | 9.8e-05 | 163.80   9928.728          |

## Parameters

| name          | value (rounded) | hesse      | at limit |
|---------------|-----------------|------------|----------|
| yield_g_jpsi  | 60.7562         | +/- 7.2    | False    |
| xs_psi2s      | 7.66673         | +/- 8.7    | False    |
| yield_bkg     | 270.41          | +/- 21     | False    |
| mu_jpsi       | 3.093           | +/- 0.0034 | False    |
| sigma_cb_jpsi | 0.0402194       | +/- 0.01   | False    |
| alpha_jpsi    | 1.32574         | +/- 0.6    | False    |
| n_jpsi        | 2.66748         | +/- 4.1    | False    |
| sigma_g_jpsi  | 0.0250128       | +/- 0.0065 | False    |
| a1            | -0.728864       | +/- 0.095  | False    |
| a2            | -0.110889       | +/- 0.12   | False    |
| a3            | 0.0537417       | +/- 0.11   | False    |

# LS run test: table of parameters and results

```
# Counting how many adjacent residuals have the same sign***
```

```
# Counting the positive and negative residuals***
```

```
Number of positive residuals: 28
```

```
Number of negative residuals: 26
```

```
Measured number of runs: 34
```

```
Expected number of runs: 28.0 with std = 3.6
```

```
#probability function of runs***
```

```
Normalization: 0.9999999999999855
```

```
#Defining the rejection interval***
```

```
Integrating up to  $R = 21$ : cumulative = 0.037
```

```
Given a significance level of 0.95, the critical region is  $R < R_{\min} = 22$ 
```